

Efficient Vision Transformer for Human-Centric AIoT Applications Through Token Tracking Assignment

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Abstract—The integration of Artificial Intelligence Internet of Things (AIoT) with consumer electronics has resulted in enhanced connectivity and intelligence within the consumer electronics sector, thereby facilitating a more convenient living experience for individuals. Particularly, the utilization of Vision Transformer (ViT) in AIoT has further improved the decision-making capabilities of consumer electronics devices. To cater to the increasing demand for efficient and intelligent consumer electronics, the prevalent approach involves transferring ViT models, which are trained on AIoT cloud servers, to edge devices for deployment. However, the incongruity between resource-constrained edge devices and computationally intensive ViT models necessitates the reduction of redundant computations in ViT to optimize the utilization of edge devices. This paper introduces a novel approach named TMTP, which combines token merging and pruning techniques. TMTP encompasses three key blocks: the Token Merging Block (TMB), the Token Tracking Assignment Block (TAB), and the Token Pruning Block (TPB). TMTP effectively diminishes redundant computations in the ViT model while preserving its accuracy. The experimental results demonstrate that by employing TMTP to hierarchically prune 80% of the input tokens in the SWAG model, the number of GFLOPs is significantly reduced by 35%. Interestingly,

the accuracy of ImageNet-1k experiences a mere reduction of 0.05%-0.18%.

Index Terms—Artificial intelligence of things (AIoT), token merging, token pruning, vision transformer.

I. INTRODUCTION

CONSUMER electronics refers to devices and equipment designed for consumer-grade usage in various fields, such as medical, transportation, and daily living [1]. However, the lack of effective interconnection among these devices hinders their convenience and intelligence in use [2]. For instance, traditional medical devices often employ different operating systems and data formats, impeding data transmission and sharing among different medical organizations and ultimately undermining the quality of medical services. Additionally, the inefficient data collection and processing capabilities of existing electronic devices restrict the utilization of vast amounts of generated information for offering personalized and customized services [3]. An illustrative example is the limited data and sluggish data processing associated with conventional traffic devices, which prevent effective analysis and utilization of extensive traffic data.

In response to the problems mentioned above and the goal of enhancing user experience, the integration of Artificial Intelligence Internet of Things (AIoT) into consumer electronics has gained significant attention [4]. This integration has made significant advancements in various fields, including smart transportation [5], [6], [7], smart vehicles [8], smart home [9], [10], [11], and smart medical [12], [13]. AIoT, which combines Artificial Intelligence (AI) with the Internet of Things (IoT) [14], [15], enables the connection of physical objects through the Internet and facilitates the real-time collection, analysis, and processing of data using AI techniques [16]. As consumer electronics continue to develop and integrate with AIoT, individuals are becoming increasingly interconnected with smart devices and technologies in their daily lives. The establishment of communication interfaces within electronic devices allows for seamless data transfer and sharing within a local area network [17]. Moreover, machine learning techniques applied in the context of AIoT enables the analysis of data collected from consumer electronic devices [18], [19]. For example, AIoT systems leveraging image recognition have been implemented for real-time pneumonia detection in children, improving the diagnostic

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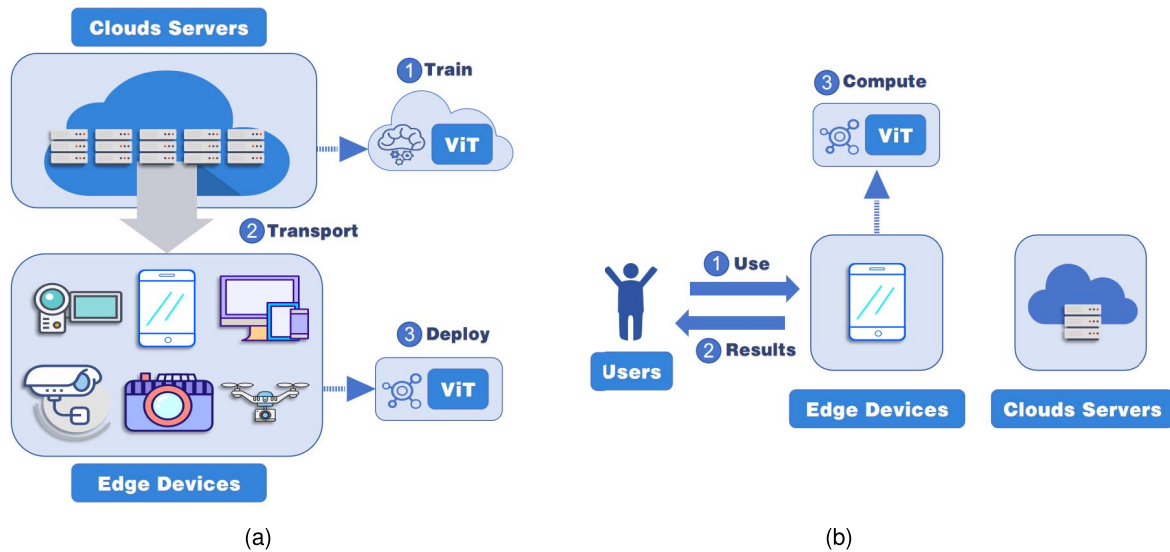


Fig. 1. Overview of the partial architecture of AIoT. (a) The process of training and deploying ViT in AIoT. (b) Image processing process based on edge devices.

process [20]. Additionally, a vehicle-networked nighttime target detection system has been proposed to provide intelligent guidance, reducing traffic congestion and accidents during nighttime hours [8]. Image recognition techniques play a crucial role in such AIoT applications, significantly enhancing people's quality of life through intelligent and efficient solutions [10]. However, traditional image recognition models still need improvement in terms of efficiency and accuracy, especially when dealing with large-scale, real-time, and diverse datasets due to their limited transfer learning capabilities and low operational efficiency. Therefore, integrating more advanced models into AIoT becomes crucial to meet the growing consumer demand for electronic devices and provide personalized services and recommendations [18].

Vision Transformer (ViT) models have shown exceptional performance in various computer vision tasks [21], such as image classification [22] and object detection [23], and have surpassed traditional models. This is particularly relevant in the context of meeting the efficiency and intelligence demands of consumer electronic products. Current mainstream practice involves training ViT models in AIoT edge/cloud servers [24], [25], [26], [27], [28], [29], [30]. Teerapittayanon et al. [26] proposed a distributed deep neural network architecture that includes cloud servers, edge servers, and AIoT devices. Similarly, Kang et al. [25] suggested partitioning the neural network model into two parts, running one part on cloud servers and the other on AIoT devices. For cloud servers and edge devices, ViT can be deployed locally for execution. As shown in Figure 1(a): ① The ViT model is trained on cloud servers. ② After training, the model is saved and transferred to the edge devices. ③ The deployed ViT model can be locally executed on the edge devices, enabling simple image processing and analysis without the need for transferring images to cloud servers. In Figure 1(b): ① Users can directly use edge devices such as mobile phones, computers, and drones. ② ViT models can

directly acquire image data and perform simple calculations and analysis. ③ Edge devices return the results of data processing to the users. This approach reduces data transmission and response time while improving the efficiency of image processing. Recognizing the remarkable capabilities of ViT, Liu et al. [31] proposed combining convolution and ViT for real-time monocular depth estimation on resource-constrained AIoT edge devices. Additionally, Lu et al. [24] presented Transmission-Friendly ViT (TFormer), which achieves highly accurate and efficient image classification on edge devices using cloud-assisted training. However, the limited computational resources of AIoT edge devices pose a challenge [32]. Moreover, ViT models have a notable disadvantage in their use of image patches as input tokens [33], [34], potentially leading to redundant computations. In the medical field, ViT models have shown potential in accurately diagnosing and planning treatments by analyzing medical data. However, the limitations of ViT models in handling large-scale medical datasets due to redundant computations hinder real-time analysis. Similarly, in the transportation field, the existence of redundant computations in ViT models hampers their mass deployment on edge devices for traffic monitoring. Resolving the issue of redundant computations in ViT models is crucial to improve the data analysis capability and resource utilization of edge devices in the context of AIoT.

Existing approaches take two paths to reduce redundant tokens in ViT models.

- 1) The first path is token pruning. Dynamic Token Sparsification ViT (DynamicViT) [33] dynamically prunes redundant tokens on a layer-by-layer basis based on input tokens. Adaptive Tokens for Efficient Vision Transformer (A-ViT) [35] is an approach that adaptively adjusts the cost of ViT inference for different images. It can automatically reduce the number of tokens that need to be processed in the network during the inference process with a lightweight prediction module. However, a

TABLE I
THE LIST OF ABBREVIATED TERMS USED IN THIS PAPER

Name	Description
AIoT	Artificial Intelligence of Things
ViT	Vision Transformer
TMTF	token merging and pruning
TMB	Token Merging Block
TPB	Token Pruning Block
TAB	Token Tracking Assignment Block
TSH	Token Size Sensing Head
TRH	Token Importance Reshaping Head
EViT	Expediting Vision Transformer
DynamicViT	Dynamic Token Sparsification ViT
A-ViT	Adaptive Tokens for Efficient Vision Transformer
PS-ViT	Patch Slimming Methods for Efficient Vision Transformer
TCFormer	Token Clustering Transformer
ToMe	Token Merging
AugReg	Data Augmentation
MAE	Masked Autoencoders
SWAG	Weakly Supervised through hashtAGs

TABLE II
THE LIST OF SYMBOLS AND NOTATIONS USED IN THIS PAPER

Name	Description
n	the number of chunks into which input tokens are divided.
N	the number of transformer layers in Vision Transformer
Q	the query matrix
q_{cls}	the query vector of class token
K	the key matrix
V	the value matrix
A	attention score
A_{cls}	class attention score
$\overline{A_{cls}}$	Multi-Head class attention score
A'_{cls}	the adjusted Multi-Head class attention score
H	the total number of attention heads
s	a row vector containing the size of each token
d	the token reduction ratio
r	the number of tokens reduced in TMB
p	the number of tokens reduced in TPB

common problem with token pruning methods is that they can result in substantial information loss if the pruning decision is not reasonable. This information loss problem limits the number of tokens that can be reduced.

- 2) The second approach is to merge tokens. Token Clustering Transformer (TCFormer) [36] can gradually aggregate and merge tokens in multiple stages, adaptively generating tokens with different positions and sizes according to different images. Token Merging (ToMe) [37] contains a lightweight matching algorithm that gradually merges similar tokens in ViT. However, the common drawback of these token merging methods is that some of the background tokens that are still retained will interfere with the final image recognition results.

By discussing the advantages and disadvantages of the token pruning approach and the token merging approach mentioned above, we can find that the two methods can complement each other, so it is feasible to combine the two methods in a technical synthesis. So we set out to find an approach that can combine the advantages of both to achieve the best results.

- 1) For the token pruning approach, although the purpose of reducing redundant computation can be achieved by directly removing redundant tokens, if the token pruning decision is not reasonable, it will directly cause a large amount of information loss. To solve the token loss problem, it is necessary to optimize the pruning decision and reduce the possibility that tokens containing valid information are mistakenly deleted.
- 2) For the token merging approach, even though the token merging avoids direct information loss, there are still some unimportant background tokens in the merged tokens, which will become interference information after

entering the attention layer and affect the final image recognition results.

To address these problems, we consider incorporating the Token Tracking Assignment Block (TAB) to connect the Token Merging Block (TMB) and the Token Pruning Block (TPB). In particular, the TAB utilizes the Token Size Sensing Head (TSH) and token Importance Reshaping Head (TRH) to adjust token importance, enabling the retention of merged tokens that contain valuable information while removing unimportant tokens. This approach balances the strengths and weaknesses of both token merging and pruning. Aside from the TAB, we suggest two additional techniques to enhance the token merging and pruning methods to reduce information loss. First, we can add a warm-up strategy to the TPB to prevent the tokens of the shallow layers from being excessively cropped and to avoid a large loss of information. Second, we can incorporate a token chunk merging operation in the TMB, which reduces the loss of local detail information in the image. Overall, our approach effectively combines token merging and pruning techniques, while introducing new strategies to mitigating information loss in the process.

In this paper, we propose a new approach that combines token merging and pruning, called TMTF. TMTF can be used to reduce redundant computations in ViT models. It consists of three blocks including the Token Tracking Assignment Block (TAB) to connect the Token Merging Block (TMB) and the Token Pruning Block (TPB). Among them, the most important TAB acts as a link between TMB and TPB. TAB tracks token size through the Token Size Sensing Head and assigns weight to token importance through TRH. TMB performs token chunking, which allows for the creation of merged tokens that capture more local information. This approach ensures that TPB can make more reasonable pruning decisions, retaining valuable tokens containing significant information.

Additionally, TPB incorporates a warm-up strategy to minimize the loss of information caused by token pruning at shallow layers. By adding the TMTP to the ViT model, a large number of redundant tokens can be reduced while minimizing the loss of information. Compared with other token merging or pruning methods, our method can reduce more redundant computations under the same conditions of reduced accuracy. Our model has higher accuracy under the same condition of reducing redundant computations. Therefore, TMTP can make ViT models easier to be deployed for use on AIoT edge devices with limited compute and storage capacity by reducing redundant computations of ViT models. This means more electronic devices can utilize ViT to easily improve the efficiency and accuracy of data processing. For example, in medical devices, ViT can speed up the analysis of massive amounts of medical data and make more accurate judgments during diagnosis and treatment, providing more precise treatment plans based on people's individual differences. In transportation devices, ViT models can reduce unnecessary information by merging and pruning tokens, and extract and retain key feature information. In this way, traffic data can be more effectively used by the model for real-time traffic prediction, traffic congestion analysis, traffic signal optimization, and other applications. This helps drivers and traffic users to make more informed decisions, thus reducing the likelihood of traffic congestion and accidents.

We summarize our contributions as follows.

- We propose a new approach for AIoT edge devices called TMTP, which combines token merging and pruning methods using TAB to reduce redundant computations in ViT models.
- We propose TMTP, which includes a design for token merging operation to capture local image details in TMB, a token size tracking mechanism to inform token importance adjustment in TAB, and a warm-up strategy to prevent information loss at shallow layers in TPB.
- We demonstrate that TMTP can effectively reduce the computational complexity without compromising the accuracy by performing experimental results on the ImageNet-1k dataset.
- We demonstrate the potential of TMTP to enhance the level of intelligence, user experience, and safety of consumer electronics products.

We conduct extensive experiments on the ImageNet-1k dataset. We apply TMTP to three popular ViT models, including AugReg [38], MAE [39], and SWAG [40]. Our approach achieves a superior trade-off between complexity and accuracy for ViT models (Section IV-E). Moreover, by comparing our approach with state-of-the-art architectures (Section IV-G) and other token pruning and merging methods (Section IV-F), our approach achieves the best results without training. Our experimental results show that TMTP is able to effectively reduce the complexity of the model without much impact on the accuracy. The advantage of TMTP in terms of the number of GFlops exemplifies our advantage in terms of the complexity of the ViT model. It makes the transmission of ViT models from cloud servers being transferred to edge devices more efficient and also suitable for deployment on resource-constrained

AIoT edge devices. This helps AIoT edge devices better adapt to people's personalized needs for consumer electronics [41], providing more efficient computation and data analytics, more secure distributed privacy protection, and higher levels of service.

II. RELATED WORKS

Vision Transformer: The transformer model has achieved success in various computer vision tasks, including image classification, target detection, and semantic segmentation. In particular, DETECTION TRANSFORMER (DETR) [23] and ViT [42] are notable applications of the transformer model in machine vision. DETR is an end-to-end target detection network based on Transformer, while ViT is the first model to use the transformer for image classification. Recent work has improved the performance of ViT models through techniques such as Data Augmentation (AugReg) [38], Masked Autoencoders (MAE) [40], and Weakly Supervised through hashtAGs (SWAG) [43]. In this paper, we integrate TMTP into AugReg, MAE, and SWAG models to further enhance the efficiency of ViT models, making them more applicable in the field of AIoT. These models were chosen due to their advanced performance in computer vision.

Token Pruning: Token pruning is a method employed to reduce the computational costs of ViT models by removing redundant tokens. We have analyzed and introduced related work on token pruning, including DynamicViT [33] and A-ViT [35], in detail in Section I. In addition to that there are a new patch slimming approach [44] and the Patch Slimming Methods of the Vision Transformer (PS-ViT) [45], and we find that they also have the same problem that they all focus on token pruning only. However, these methods often result in significant information loss due to unreasonable pruning decisions, limiting the number of tokens that can be reduced. Our approach aims to overcome this limitation by optimizing token pruning decisions using size information and implementing a warm-up strategy to gradually increase the number of pruned tokens at each layer, thus minimizing information loss.

Token Merging: In addition to token pruning, token merging is another technique utilized to reduce redundant computations in the ViT model. Existing methods that we analyzed in Section 1 include TCFormer [36] and ToMe [37]. In addition to these there is EViT [34]. However, these approaches retain unimportant background tokens in the merged tokens, which can interfere with image classification results. In contrast, our approach incorporates token pruning after token merging to eliminate the unimportant background tokens and introduces the chunking operation during the merging process to capture more local detail information. By doing so, our approach reduces redundant computations while preserving local detail information and improving model accuracy.

III. METHOD

A. Overview Architecture

The overall structure of our TMTP is presented in Fig. 2, which includes three blocks: TMB, TAB, and TPB. Among them, in TMB, tokens are first subjected to a chunking

operation and then merged within their chunks. Finally, all the chunks are merged into one. In TAB, token importance is adjusted based on the size of the merged token. The TSH is used to track the size of the token, and the TRH assigns weights to the token importance $\overline{A}_{cls}$ based on the token size. In TPB, unimportant tokens are pruned based on token importance. TIH is a Token Identification Head to identify insignificant background tokens. TMTP is added between the attention module and the mlp module in the ViT model.

We first introduce the concept of token similarity and the meaning of token chunking merging in the TMB. Then, the concept of token importance in TAB, and the process of adjusting token importance are introduced. Finally, we explain the working principle of the Token Identification Head in the TPB and the significance of the warm-up strategy.

B. Token Merging Block

1) Token Similarity Before merging similar tokens in TMB, we first need to define the concept of token similarity. One way to gauge similarity between tokens is to measure the distance between their features [37]. However, this approach may not always be the most effective one. The ViT models [42] such as ViT-S, ViT-B and ViT-L have token vector spaces of dimensions 384, 768 and 1024. These over-parameterized intermediate feature spaces contain some irrelevant noise, leading to the generation of erroneous cross-token similarity measures. To accomplish tokenization, we separate the input picture into patches, each of which is projected to the corresponding token embedding. The Multi-Head Self-Attention (MHSA) mechanism then processes these tokens at the attention layer to provide an answer, as shown in Eq. (1) below:

$$A(Q, K, V) = \text{Softmax}\left(\frac{Q \cdot K^T}{\sqrt{C}}\right)V \quad (1)$$

In this equation, the token's embedding dimension is C , and the query, key, and value vectors are Q , K , and V , respectively. The K represents the attention weight of each token in the input sequence, which in turn indicates the importance or similarity of the feature vector represented by each token. So we use the dot product similarity between the K values of all tokens to represent the similarity between tokens. After that, we use the bipartite soft matching approach to merge the tokens with high similarity according to the similarity between tokens [37]. The number of tokens that will be reduced by token merging in each block of the transformer is r .

2) Merging Tokens in Chunks We propose to divide input tokens $X = [x_1, x_2, \dots, x_k] \in \mathbb{R}^{B \times N \times C}$ into n large chunks, and then perform token merging within each chunk separately. To distribute the data stream across n processes, we split X into n equal parts using the formula $X_i = [x_1, x_2, \dots, x_{\frac{k}{n}}] \in \mathbb{R}^{B \times \frac{N}{n} \times C}$. Then, we perform token merging within each block separately using the matching approach. This approach offers several advantages. Firstly, it reduces the computational cost of the model. In the absence of chunking, the total number of matches that need to be performed by the tag using the matching approach for merging is $N \times N$ times. However, chunking reduces the total number of required matching operations

to $n \times \frac{N}{n} \times \frac{N}{n}$ times. The number of matching operations is inversely proportional to the number of chunks n , the number of matching operations is reduced to $\frac{1}{n}$ times the original number. Secondly, chunking enables better capture of the local structure within the input. By breaking the input data into multiple chunks, a hierarchical structure is created that is akin to clauses and paragraphs in natural language [46], [47]. Chunking facilitates the division of lengthy text into concise and meaningful parts, aiding in natural language processing. Likewise, chunking allows us to take advantage of the local structure within each chunk, thereby preventing the merging of tokens across the entire image and losing local information.

C. Token Tracking Assignment Block

1) Token Importance Before adjusting the importance of tokens in the TAB, we first need to define the concept of importance. We define the class token attention score of a token as the token importance. So what is the class token attention score? Suppose the input sequence of ViT encoder consists of n tokens, and each token includes a class token. The class token will be used separately in the final classification process to determine the class of the input image. The interaction between class tokens and other tokens is carried out through the attention mechanism in ViT:

$$A_{cls} = \text{Softmax}\left(\frac{q_{cls} \cdot K^T}{\sqrt{C}}\right) \quad (2)$$

where q_{cls} , K and V represent the query vector, key matrix, and value matrix of class token. The class token attention score, A_{cls} , represents the correlation between each token and the class token. The higher the class token attention score of a token, the stronger the correlation between the token and the class token, indicating its importance for the final classification result. Therefore, we propose to use the class A class of the token attention score to represent the token importance. The calculation of Eq. (2) in ViT, which is performed simultaneously in multiple heads, is called the MHSA mechanism:

$$\overline{A}_{cls} = \sum_{h=1}^H A_{cls}^{(h)} / H \quad (3)$$

In Eq. (3), there exist multiple class token attention vectors: $A_{cls}^{(h)}$, $h = [1, 2, \dots, H]$, with H denoting the overall number of attention heads. Therefore, when the total number of attention heads is greater than 1, we define the token importance as the average class token attention score $\overline{A}_{cls}$.

2) Tracking Token size Tokens no longer represent a single input patch after they are merged in the TMB. If these merged tokens are cropped or ignored, the accuracy and generalization ability of the model will be affected because they cover the information of numerous tokens. Therefore, we need to increase the model's attention to this information and reduce the possibility of them being cropped. Furthermore, the token importance will be used in the subsequent TPB as a criterion to determine the importance of a token. So in the TAB, we need to adjust $\overline{A}_{cls}$ according to the size of the merged token. In TAB, the token size is tracked by the TSH, and then the TRH

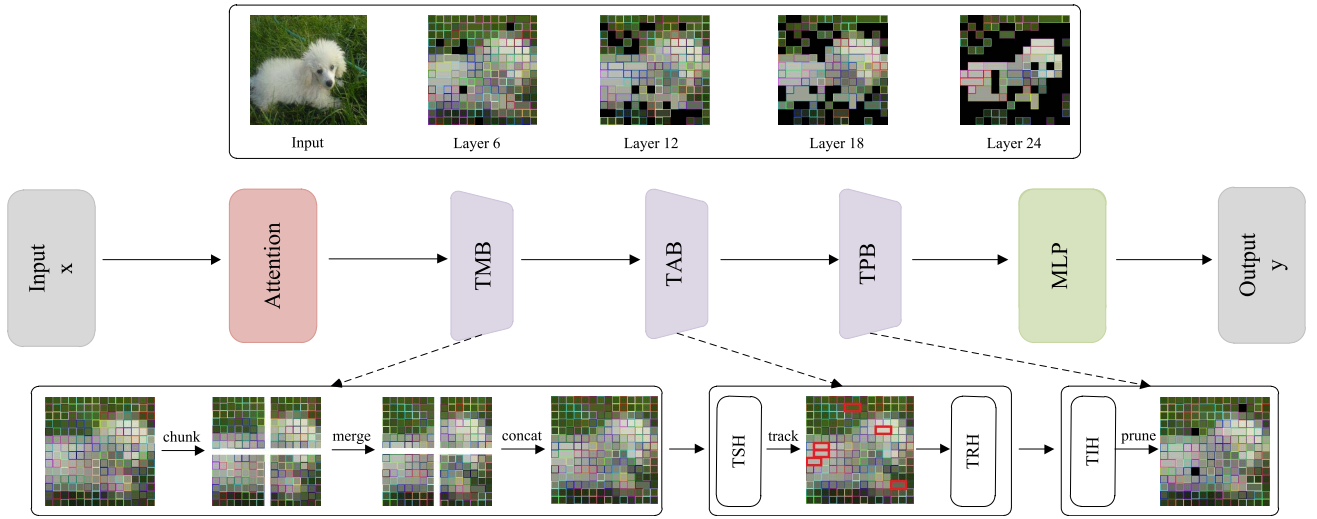


Fig. 2. Overview of TMTP.

is used to assign weights to $\overline{A}_{cls}$ according to the merged token size. The specific process is shown in the following equation:

$$A'_{cls} = \overline{A}_{cls} + \log s \quad (4)$$

where s is a row vector containing the sizes of each token and A'_{cls} represents the adjusted Multi-Head class attention score. In this way, the token importance can be adjusted in the TAB based on the size information of the merged tokens to complete the adjustment, making the individual token importance scores more reasonable.

D. Token Pruning Block

1) Pruning Background Token At the beginning of the Token Pruning Block TPB, we set a Token Identification Head (TIH), as shown in Fig. 2, to identify unimportant background tokens. In TIH, we use top- k to select the p token with the smallest value of token importance A'_{cls} , which we call the unimportant background token, and the rest are the important body tokens [48]. Since we assigned different weights to A'_{cls} based on token size in the previous TAB, the more informative merged tokens will have larger A'_{cls} values. So most of the p tokens selected make unimportant background tokens that can be removed without significantly affecting the performance of the ViT model. The remaining more important subject tokens will be retained.

2) Increasing The Number of Pruning Previous research has shown that the correlation of tokens in the shallow transformer layer is low, making it difficult to distinguish subject tokens from background tokens in these layers. Additionally, tokens in the shallow and intermediate layers typically have rapidly growing feature representation capabilities. Therefore, In the shallow layers of the visual transducer, it is important to avoid pruning too many tokens, as this can lead to substantial information loss. To optimize the performance of ViT models, we use a preheating strategy to determine the appropriate number of pruning tokens per layer. Specifically, the number of tokens that prune p in each layer is gradually increased as the number of transformation layers advances. With the

preheating strategy, we can reduce the loss of tokens in the shallow layers [49], maintain fast inference speed and high model efficiency, and achieve a trade-off between information loss and model efficiency.

IV. EXPERIMENTS

A. Setup

We base TMTP on three ViT models (*i.e.*, AugReg [38], MAE [40], and SWAG [43]) trained in three different ways. We evaluated our image classification approach on a large-scale 1000-class ImageNet-1k [50] dataset with a pixel resolution of 224×224 . The training set of ImageNet-1k contains 1200k images, the validation set contains 50k images and the test set contains 100k images. In our experiments, we apply TMTP to some off-the-shelf ViT models and evaluate them on ImageNet-1k val. For MAE models, the TMTP will further be trained after applying it to the model. Unless otherwise stated, all throughputs are measured while performing inference on a V100 GPU with optimal batch size and fp32.

B. Architecture Design Details

We evaluate three different types of models including MAE [40] (ViT-H, ViT-L, ViT-B), SWAG [43] (ViT-H @ 518, ViT-L @ 512, ViT-B @ 384), and AugReg [38] (ViT-L, ViT-B, ViT-S, ViT-T) models on the ImageNet-1k validation set. all of these models above have different numbers of layers and input sizes. these models above have different numbers of layers and input sizes. For MAE and AugReg models, due to the relatively small input size, resulting in a relatively small total number of tokens, we set to insert in the each layer in ViT-B, ViT-S and ViT-T the TMTP, n and r values are kept constant, and p value is increased layer by layer using the warm-up strategy; in ViT-L and ViT-H, due to the relatively large number of layers, we choose to insert TMTP in each of the even numbered layer, n and r values are kept constant, and p value is increased layer by layer using the warm-up strategy. In the SWAG model, the input size value is relatively large, resulting

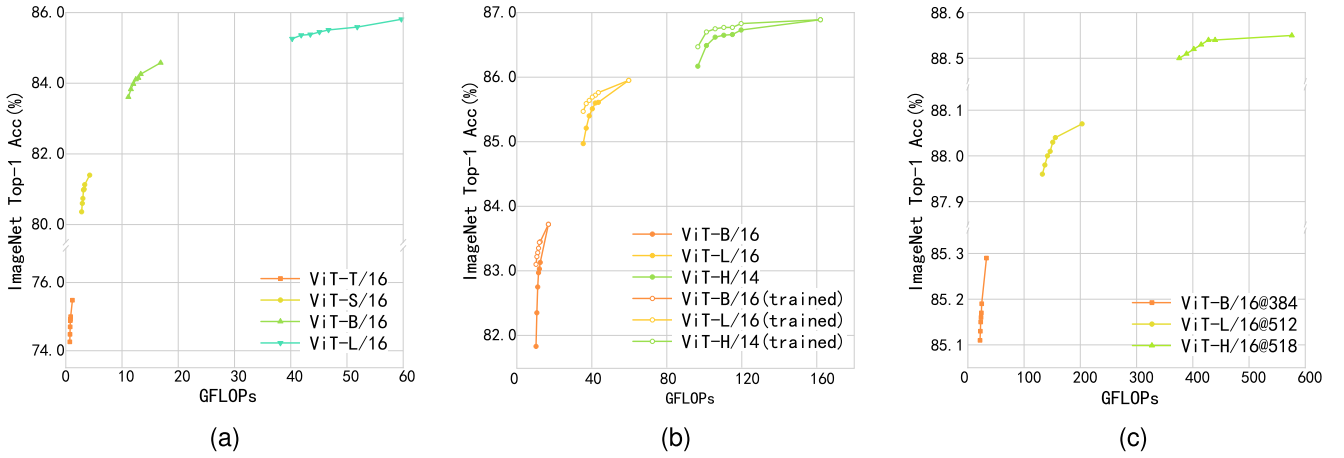


Fig. 3. Main results of applying TMTP to state-of-the-art ViT models. (a) AugReg Models. (b) MAE Models. (c) SWAG Models.

in a large total number of tokens, so we set to insert TMTP in each layer, n and r values are kept constant, and p values are increased layer by layer using the warm-up strategy with a relatively large growth rate. We refer to the notion of token keeping ratio in DynamicViT [33] and define the token reduction ratio as d , which denotes the ratio of reduction tokens to total input tokens. The n and r values are fixed for each model, after which the token reduction ratio d can be changed accordingly just by adjusting the size of the p value.

C. Evaluating and Fine-Tuning Details

For the AugReg and SWAG models, which are pre-trained on large supervised pre-training datasets, we apply TMTP to them and evaluate them in ImageNet-1k val. For the self-supervised model MAE model, in addition to evaluating it in ImageNet-1k val, we will train it using the method of fine-tuning public pre-training checkpoints.

Hyperparameter optimization is usually performed during the training process of the model [51]. Hyperparameters such as learning rate, batch size, regularization parameters in the model [52], [53] have been tuned in MAE [40] and the best model performance has been found. Since our approach can be easily incorporated into these models without substantially modifying them, the original training formulation works well with our approach. Therefore, as with EViT [34] and ToMe [37], we refer to the training equations in the original MAE [40] paper.

D. Using Token Tracking Assignment Block

To combine the advantages of both token merging and pruning methods, we propose to use a TAB to connect the TMB with the TPB. To evaluate our approach, we compare the performance of a MAE model (ViT-L) with and without the TAB. Our findings presented in Tab. III and Tab. IV. As can be seen from the table, compared to the model without the TAB, the models with the TAB produce a higher accuracy. Furthermore, we observed that using ViT-L showed to produce less volatility in the top-1 accuracy of the model than the same baseline. Significantly, we note that the volatility difference between the two models increases as the token reduction ratio d increases. Overall, the inclusion of the TAB in our ViT

TABLE III
ViT-L WITH TOKEN TRACKING ASSIGNMENT BLOCK

Dropping Ratio d	Top-1 Acc(%)	GFLOPs
with token tracking assignment block		
0	85.95	59.69
0.65	85.61(-0.39%)	43.49
0.70	85.60(-0.40%)	41.87
0.75	85.51(-0.51%)	40.26
0.80	85.40(-0.63%)	38.64
0.85	85.21(-0.86%)	37.03
0.90	84.97(-1.10%)	35.41

TABLE IV
ViT-L WITHOUT TOKEN TRACKING ASSIGNMENT BLOCK

Dropping Ratio d	Top-1 Acc(%)	GFLOPs
without token tracking assignment block		
0	85.95	59.69
0.65	85.59(-0.42%)	43.49
0.70	85.56(-0.45%)	41.87
0.75	85.5(-0.52%)	40.26
0.80	85.36(-0.69%)	38.64
0.85	85.15(-0.93%)	37.03
0.90	84.82(-1.31%)	35.41

model improved the model's robustness, resulting in minimal accuracy loss.

E. Main Results

One advantage of TMTP is that it can be applied to ViT of various architectures. As shown in Fig. 3, we summarise the main results on ImageNet-1k, where we evaluate TMTP using three classes of basic models (AugReg, MAE, SWAG). TMTP is robust to token reduction and achieves high accuracy at different token reduction rates d .

As shown in Fig. 3, TMTP compress 80-90% of the tokens in the model, reducing the computational cost of the model by 35% to 39% with negligible impact on performance. Even in the case of 80% token reduction, the impact on accuracy

TABLE V
COMPARISON TO TOKEN PRUNING, TOKEN MERGING AND
STATE OF THE ART AT “SMALL” MODEL COMPLEXITY

Model	Input	GFLOPs	Top-1 Acc(%)
ours			
ViT-S ^{AugReg} _{d=80%}	224	2.6	79.5
SP-ViT	224	2.6	-
ToMeViT-S ^{AugReg} _{r13}	224	2.7	79.3
DynamicViT	224	2.9	79.3
A-ViT	224	0.9	-
ViT-S ^{AugReg}	224	4.2	81.4
LV-ViT-S	224	6.6	83.0

TABLE VI
COMPARISON TO TOKEN PRUNING, TOKEN MERGING AND
STATE OF THE ART AT “BASE” MODEL COMPLEXITY

Model	Input	GFLOPs	Top-1 Acc(%)
ours			
ViT-B ^{AugReg} _{d=90%}	224	10.2	83.1
ViT-B ^{AugReg} _{d=85%}	224	10.7	83.4
ToMeViT-B ^{AugReg} _{r13}	224	10.1	82.6
ToMeViT-B ^{AugReg} _{r12}	224	10.7	82.9
LV-ViT-M	224	12.7	84.0
ViT-B ^{AugReg}	224	16.9	84.6
CSWin-B	224	15.0	84.2
Swin-B	224	15.4	83.3
Swin-B*	224	15.4	84.0
Eff-B6	528	19.0	84.0

in the AugReg model is small (−0.54% to −1.22%). In the SWAG model, the impact on accuracy is even more negligible (−0.05% to −0.18%). We believe that this trend is not just because SWAG models have a higher number of tokens, but because these larger models have more layers, allowing for more gradual changes in functionality as the model gets deeper, thus reducing the negative impact. For the MAE model, the computational cost is reduced by 40% when using an off-the-shelf model directly, while the accuracy only decreases (−0.72% to −1.89%). After training, the accuracy is even improved compared to the pre-training (−0.42% to −0.70%). Not only do we see the same trend in the MAE model as before, but we also find that we can reduce the computational costs by reducing the number of tokens in the fine-tuning, which improves the training speed and therefore allows us to train larger models than before. These favorable results demonstrate the potential of TMTP to reduce computational costs, enhance training speed, and enable the development of larger models, providing consumers with better, more convenient, and more intelligent services in the realm of consumer electronics.

F. Comparison to Token Merging and Pruning

In Tab. V, Tab. VI, Tab. VII and Tab. VIII, d denotes the token reduction ratio. * denotes the model use SimMIM self-supervised pretraining. **Pink** indicates the model that uses TMTP. *trained* indicates TMTP is applied during training. We first compare the TMTP with ToMe in four different sizes of model complexity levels. We first compare the model performance after applying two different methods for training. In the case of training ViT-H using the

TABLE VII
COMPARISON TO TOKEN PRUNING, TOKEN MERGING AND
STATE OF THE ART AT “LARGE” MODEL COMPLEXITY

Model	Input	GFLOPs	Top-1 Acc(%)
ours			
ViT-L ^{MAE} _{d=90%}	224	36.7	85.1
ViT-L ^{MAE} _{d=90%} (<i>trained</i>)	224	36.7	85.5
ToMeViT-L ^{MAE} _{r6}	224	37.6	85.1
ToMeViT-L ^{MAE} _{r6} (<i>trained</i>)	224	37.6	85.4
ViT-L ^{MAE}	224	61.6	85.7
MViTv2-L	224	42.1	85.3

TABLE VIII
COMPARISON TO TOKEN PRUNING, TOKEN MERGING AND
STATE OF THE ART AT “HUGE” MODEL COMPLEXITY

Model	Input	GFLOPs	Top-1 Acc(%)
ours			
ViT-H ^{MAE} _{d=85%}	224	100.2	86.5
ViT-H ^{MAE} _{d=85%} (<i>trained</i>)	224	100.2	86.7
ToMeViT-H ^{MAE} _{r6}	224	100.9	86.3
ToMeViT-H ^{MAE} _{r6} (<i>trained</i>)	224	100.9	86.5
ViT-H ^{MAE}	224	167.4	86.9
Swin-V2-H*	224	118.1	85.7

same fine-tuning strategy, TMTP results in a greater decrease in model complexity and higher model accuracy. Second, when directly applied to off-the-shelf models, TMTP performs better than ToMe in ViT-S, ViT-B, ViT-L, and ViT-H. Specifically, when the GFlops of ViT-B are all reduced by about 40%, TMTP results in less decrease in the accuracy of the model. When the accuracy of the ViT-L models are all approximately reduced by about 0.7%, TMTP can reduce the GFlops of the models by more. These results suggest that our method has a better model complexity-accuracy trade-off compared to ToMe. In Tab. V, We then compare TMTP with a variety of token pruning and token merging methods also applied in ViT-S, including newer methods such as A-ViT [35], DynamicViT [33], SP-ViT [54], and ToMe [37]. Without training, TMTP already performs better in reducing model complexity and maintaining accuracy compared to other methods. Comparison with these state-of-the-art token merging or pruning methods further validates the advantages of combining token merging and pruning methods. Therefore, our experimental results provide strong validation that TMTP can effectively enhance the human-centric nature of consumer electronics in comparison to other existing methods. As a result, TMTP can make ViT easier to be deployed with AIoT edge devices than other existing approaches, thus expanding the scenarios in which people use consumer electronics.

G. Comparison to State of the Art

TMTP was applied to the AugReg models (ViT-S, ViT-B) and MAE models (ViT-L, ViT-H), which have complexity levels of “base”, “large”, and “huge”, respectively. As shown in Tab. V, Tab. VI, Tab. VII and Tab. VIII, we compared the models after applying TMTP with other models trained on ImageNet-1k (e.g., Swin [55], SwinV2 [56], CSWin [57], MViTv2 [58], EfficientNet [59], [60], [61] and LV-ViT [62]).

In all four complexity levels, TMTP achieves a balance between model performance and complexity reduction and is more effective when the model is trained. For example,

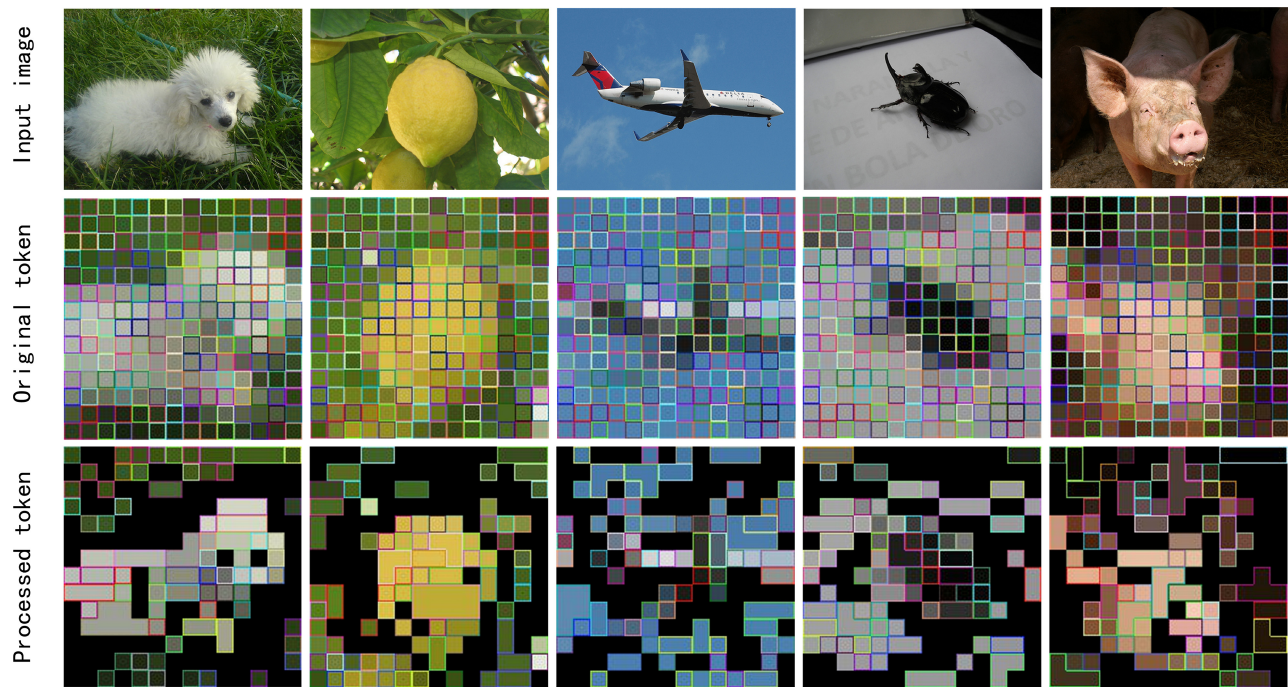


Fig. 4. Visualization results of TMTP on ImageNet-1k dataset.

the MAE models (ViT-L, ViT-H) with TMTP applied and trained have higher model accuracy and lower model complexity compared to MViTv2-L and Swin-V2-H*, respectively. In summary, TMTP is very competitive among visual models in terms of computational accuracy trade-offs. Experiments performed on different models and complexity levels demonstrate the robustness and generalisation ability of TMTP. These experimental results show that TMTP provides a better trade-off between model complexity and accuracy than other approaches. Therefore, by using TMTP, ViT can be more efficient and accurate, and thus more responsive to user needs and preferences. These results highlight the superior performance of TMTP in making consumer electronics more efficient and accurate, ultimately catering to the needs and preferences of users.

H. Visualization

To validate the effectiveness of TMTP, we applied TMTP to the AugReg model (ViT-L) on the ImageNet-1k dataset. a total of 80% of the markers were reduced. As shown in Fig. 4, TMTP effectively merges some of the subject-tagged tokens while pruning the unimportant background tokens. We observe that most of the merged tokens belong to the subject tokens and retain the important information. The first image illustrates the preservation of the toy poodle's head and body parts while removing the background, whereas the second image shows the lemon being preserved and the background branches being pruned. In the third image, the main part of the airplane is kept, with the wing and fuselage being clearly visible. Remarkably, the small patch representing the red logo in the tail of the airplane is retained, demonstrating our approach's fine-grained pruning. In the fourth image, the tentacle part of

the rhinoceros beetle is preserved, which belongs to a different token than the armor body part. Finally, the fifth image depicts the preservation of the pig's head, which contains the most critical information.

Our TMTP approach outperforms traditional merging or pruning methods in reducing the number of tokens by 75% and pruning a considerable number of background tokens while most of the important subject tokens, retaining most of the valid information. In summary, our approach achieves a balance between retention of important information and reduction of redundant tokens.

V. CONCLUSION

In this paper, we present a novel approach called TMTP for AIoT edge devices, specifically targeting the enhancement of Vision Transformer (ViT) models. Our aim is to address the increasing need for efficient computational strategies in the era of human-centric consumer electronics. TMTP integrates token merging and pruning techniques to reduce redundant computations in ViT models, enabling edge devices to process visual information more efficiently. We conducted experiments using the widely recognized ImageNet-1k dataset and found that our TMTP approach remarkably reduces the redundant computations of ViT models without compromising their accuracy. This reduction in computational complexity facilitates minimizes the computational resources consumed by edge devices.

In our future work, we plan to further explore the practical applications of TMTP in various consumer electronic products. For example, we aim to implement TMTP in smart street lights in smart cities, leveraging its capabilities to enhance people and vehicle monitoring, thus enabling greater data volume. Additionally, we envision integrating TMTP into

health monitoring devices in smart medical to optimize the recognition of vital health indicators such as skin condition, body temperature, and heart rate. Moreover, we propose integrating TMTP into driver assistance systems in smart vehicles to enable real-time monitoring of driver fatigue. Our approach will contribute to the advancement of intelligence, user experience, and safety in consumer electronics products. By utilizing TMTP, consumer electronic devices will become more intelligent, offering a higher level of convenience and providing more human-centric services to individuals in various aspects of their lives.

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